DANMARKS TEKNISKE UNIVERSITET
Institute of Electrical and Photonic Engineering

BACHELOR PROJECT PLAN

107210

Design of long-range safety control for autonomous robots

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ECTS:	20

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1 Project Description

English

For outdoor autonomous mobile robots, the current solution relying on a gamepad is too limited. This project aims to develop a long-range wireless remote control system with safety features for autonomous robots. The system must detect loss of communication and cut power to motors, allow manual override of velocity and turn rate, and operate for at least 8 hours on battery. Optionally, the remote should display messages from the robot such as battery status.

The project involves circuit and PCB design, as well as 3D-printed casing for the devices.

Dansk

For autonome mobile udendørs robotter er den nuværende løsning med en gamepad for begrænset. Dette projekt har til formål at udvikle et langtrækkende trådløst fjernbetjeningssystem med sikkerhedsfunktioner til autonome robotter. Systemet skal detektere tab af kommunikation og slukke for strømmen til motorerne, tillade manuel overstyring af hastighed og svinghastighed og fungere i mindst 8 timer på batteri. Fjernbetjeningen skal eventuelt vise beskeder fra robotten, såsom batteristatus.

Projektet involverer kredsløbs- og PCB-design samt 3D-printet kabinet til enhederne.

2 Project Scope

The project scope includes the following requirements:

1. **Communication Loss Detection:** System must detect when communication is lost and immediately cut power to the robot's motors.
2. **Manual Control Override:** The remote must allow the operator to take over control of the vehicle, controlling drive velocity and turn rate.
3. **Power Supply - Robot Side:** The mobile robot shall power the on-board receiver/control unit.
4. **Power Supply - Remote Side:** The hand-held remote shall be battery powered with at least 8 hours of operation on a fully charged battery.
5. **Status Display (Optional):** The remote end should be able to display messages from the robot, e.g., remaining battery power.
6. **Communication Frequency:** The system shall use the 433 MHz band for communication.
7. **Hardware Development:** The project shall include circuit and PCB design for both the remote and robot units.
8. **Enclosure Design:** 3D-printed casing shall be designed and produced for both devices.

3 Relation to Learning Objectives

Ability to work independently and structure a larger project, including following a schedule and organizing the work
The project is structured using a Gantt chart that divides the work into several phases, including research, design, implementation, testing, and documentation. This structure ensures that the project progresses systematically and that the different tasks are completed within the planned timeframe.

Ability to summarize and interpret technical information and perform technical problem solving through project work
During the project, technical documentation such as datasheets for radio modules, and other electronic components is studied and interpreted. This information is used to design the system and solve technical challenges.

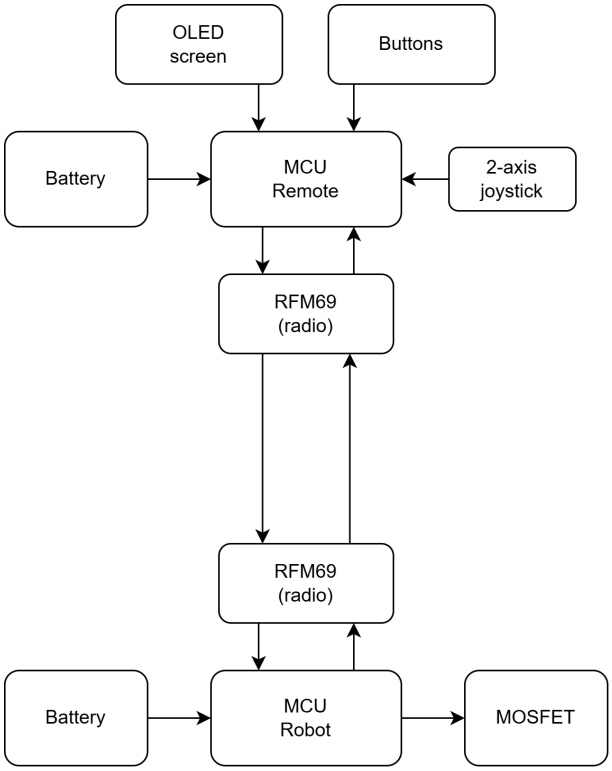
Ability to work with all phases in a project, including proposal, solution development, and documentation
The project includes all major phases of an engineering project. It begins with research and planning, followed by design and prototyping, implementation of hardware and software, and finally testing and validation. The full process and results are documented in the final report.

Ability to independently acquire new knowledge and critically evaluate information
Several technologies used in the project require independent learning, including wireless communication and embedded programming. The technical sources and documentation are evaluated to ensure that appropriate design choices are made.

Ability to communicate technical information, theory, and results in written, visual, and oral form
The final report will be written continuously throughout the project in order to allow sufficient time for clear explanations and documentation. Figures will be included to visually illustrate the system and the development process.

4 Project explanation

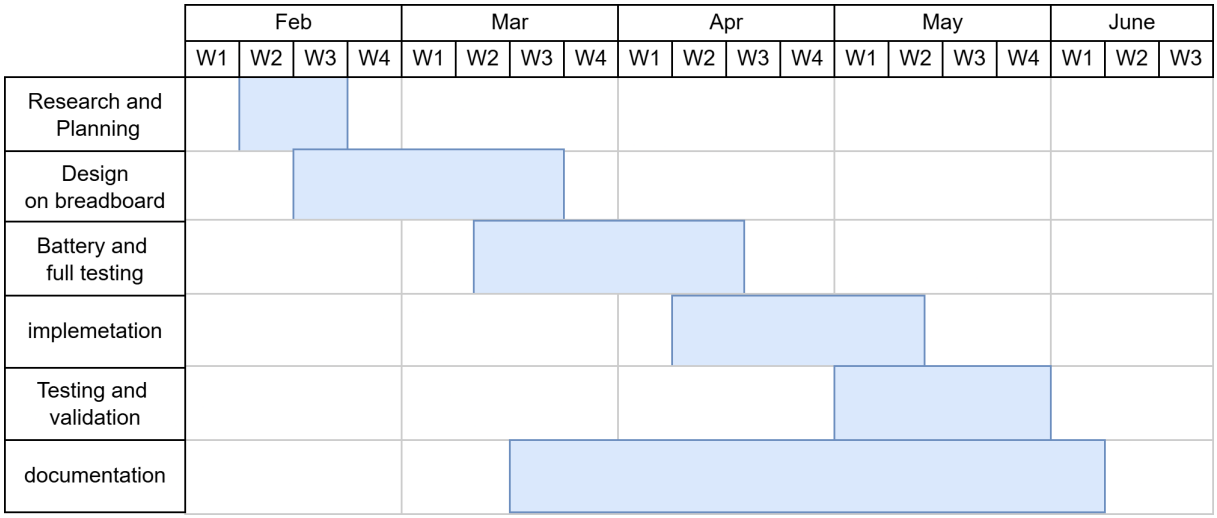
The following flowchart presents a simplified overview of how the remote control system operates. The system is divided into two main parts, each controlled by its own microcontroller (MCU). One MCU is located on the remote unit, where it handles user inputs such as the joystick and buttons and manages the display. The second MCU is located on the robot side, where it receives the transmitted signals and controls the robot's hardware components.



Figur 1: Flow chart

5 Time Schedule

The grantt chart below shows the timeline which I will be trying to follow for the bachelors project.



Figur 2: Grantt chart

6 Project Plan Description

Research and Planning
This phase focuses on understanding the problem and identifying suitable hardware and software solutions. Datasheets and technical documentation are reviewed to determine which components are most appropriate.
Risk: Selecting unsuitable components or underestimating system requirements could lead to redesign later in the project.

Design on Breadboard
In this phase, the electronic components are connected and tested on a breadboard to verify the basic functionality of the system before building a final prototype.
Risk: Hardware compatibility issues or unstable breadboard connections may delay development.

Battery and Full Testing
Power consumption and battery performance are evaluated to ensure that the system can operate reliably. Different configurations may be tested to determine the most suitable power solution.
Risk: The battery may not provide sufficient current or operating time, which may require changes to the design.

Implementation
During this phase, the final hardware is developed by designing and assembling the PCB and integrating the electronics into a 3D printed enclosure. The system is moved from the breadboard prototype to the final physical design.
Risk: Incorrect PCB design will slow project drastically and new problems might occur implementing the whole system.

Testing and Validation

The complete system is tested to verify that all components work together as intended. Communication reliability, control responsiveness, and system stability are evaluated.
Risk: Unexpected bugs or unstable communication could require additional development time.

Documentation

The final phase focuses on documenting the design, implementation, and results of the project. This includes writing the report and preparing figures and explanations.
Risk: Documentation may take longer than expected if it is postponed until late in the project.

7 References

Litteratur

[1] DTU - Technical University of Denmark. *Department of Electrical and Photonic Engineering.* Available at: `https://www.dtu.dk/english/education/undergraduate/undergraduate-programmes-in-danish/bsc-eng-programmes/electrical-engineering`